Stefan M Arseneau

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My research focuses on precision measurements of white dwarf stellar parameters and their use in stellar structure inference. I am especially interested in using gravitational redshift measurements and asteroseismology to constrain the core composition of white dwarfs at all masses, with the goal of constraining fundamental nuclear astrophysics.

Education

Boston University — PhD, MA in Astronomy — Expected 2029

Johns Hopkins University — BS in Physics, BA in Mathematics — 2024

Experience

Research Fellow, Department of Astronomy, Boston University — Boston, MA, USA — 2024 — Present

- Performed observational measurements of the white dwarf mass-radius relation and stellar structure using gravitational redshifts and asteroseismology.

Member, Milky Way Mapper, Sloan Digital Sky Survey V — 2024 — Present

- Developed and validated white dwarf parameter estimation for `astra`, Milky Way Mapper data reduction pipeline.

Research Fellow, CLASS Telescope, Johns Hopkins University — Baltimore, MD, USA — 2022 — 2024

- Modeled effects of radiative heat transfer on performance of filters using COMSOL Multiphysics.

Talks & Conferences

- Conference Organizer — Boston Area Planetary Science Meeting — Spring 2025 & Fall 2025
- Poster Presentation — SDSS-V Collaboration Meeting (MPIA, Heidelberg, DE) — 2025

  Resolution-Corrected White Dwarf Gravitational Redshifts Validate SDSS-V Wavelength Calibration and Enable Accurate Mass-Radius Tests

Observatory Allocations

- ESO Very Large Telescope, UVES, 17.5 Hours (co-PI with Roberto Raddi) — 2024

  Constraints on the Gravitational Redshifts and Mass-Radius Relation of High-Mass White Dwarfs in Wide Binaries
- Gemini Observatory, GMOS, 2 Hours — 2023

  Probing the Mass-Radius Relation of White Dwarfs in Wide Binaries

Honors

- Graduate Research Fellowship, Boston University — 2025
- Provost's Undergraduate Research Award, Johns Hopkins University — 2023

Open Source Projects

Compact Object Radial Velocities — github.com/vedantchandra/corv

- Primary author and maintainer of a scientific software package for precision radial velocity estimation.
- Integrated into the Sloan Digital Sky Survey data reduction pipeline, processing >200 GB per night.

Publications

Publications as First Author —

4. **S. M. Arseneau**; J.J. Hermes; V. Chandra; R. Raddi; M.E. Camisassa; A. Rebassa-Mansergas; S. Torres, 2026, *An Ultramassive White Dwarf with a Likely Oxygen-Neon Core*, The Astrophysical Journal, in press (arXiv:2606.19441)
3. **S. M. Arseneau**; J.J. Hermes; M.E. Camisassa; R. Raddi; E.B. Bauer, 2026, *Constraints on White Dwarf Hydrogen Layer Masses Using Gravitational Redshifts*, The Astrophysical Journal, 1000 297 (arXiv:2603.02314) [ADS Link].
2. **S. M. Arseneau**; J.J. Hermes; N.L. Zakamska; K. El-Badry; N.R. Crumpler; V. Chandra; G. Adamane Pallathadka; C. Badenes; B.T. Gänischke; N. Gentile Fusillo, 2025, *Resolution-Corrected White Dwarf Gravitational Redshifts Validate SDSS-V Wavelength Calibration and Enable Accurate Mass-Radius Tests*, The Astrophysical Journal, 991 190 (arXiv:2508.04775) [ADS Link]
1. **S. M. Arseneau**; V. Chandra; H.C. Hwang; N.L. Zakamska; G.A. Pallathadka; N.R. Crumpler; J.J. Hermes; K. El-Badry; H-W. Rix; K.G. Stassun; B.T. Gänsicke; J.R. Brownstein; S. Morrison, 2025, *Measuring the Mass–Radius Relation of White Dwarfs Using Wide Binaries*, The Astrophysical Journal, 963, 17 (arXiv:2310.19866) [ADS Link]

Publications as Contributing Author —

10. G.A. Pallathadka; V. Chandra; N.L. Zakamska; N.R. Crumpler; **S. M. Arseneau**; K. El-Badry; B.T. Gäniscke; Y. Zentai; J. J. Hermes; A.D. Schwope; C. Badenes; N.P. Gentile Fusillo; S. Morrison; T. Cunningham; P. Chakraborty; G Tovmasian; D Bizyaev; K. Pan; S.F. Anderson; S. Demasi, 2025, *Double White Dwarf Binaries in SDSS-V DR19 : A catalog of DA white dwarf binaries and constraints on the binary population,* The Astrophysical Journal, In Press (arXiv:2509.02906) [ADS Link]
9. K.R. Helson; C.Y.Y. Chan; **S. Arseneau**; A. Barlis; C.L. Bennett; T.M. Essinger-Hileman; H. Guo; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J. Wollack, 2026, *Diamond-loaded polyimide aerogel scattering filters and their applications in astrophysical and planetary science observations,* Review of Scientific Instruments, 97, 044502 (arXiv:2508.20406) [ADS Link]
8. N.R. Crumpler; V. Chandra; N.L. Zakamska; G.A. Pallathadka; **S. Arseneau**; N.P. Gentile Fusillo; J. J. Hermes; C. Badenes; P. Chakraborty; B.T. Gänsicke; S. Morrison; H-W. Rix; S.P. Schmidt; A. Schwope; K.G. Stassun, 2025, *A Large Catalog of DA White Dwarf Characteristics Using SDSS and Gaia Observations,* The Astrophysical Journal, 989 24 (arXiv:2508.00818) [ADS Link]
7. G.A. Pallathadka; V. Chandra; B.T. Gänsicke; N.L. Zakamska; D. Koester; Y. Zentai; N.R. Crumpler; **S. M. Arseneau**; J.J. Hermes; M.R. Schreiber; K.G. Stassun; A. Schwope; K. El-Badry; G. Tovmassian; T. Cunningham; S. Morrison, 2025, *Double White Dwarf Binaries in SDSS-V DR19 : The discovery of a rare DA+DQ white dwarf binary with 31 hour orbital period,* The Astrophysical Journal, In Press (arXiv:2507.11618) [ADS Link]
6. J.A. Kollmeier; H.-W. Rix; C. Aerts; J. Aird; P.V. Alfaro; A. Almeida; S.F. Anderson; O.J. Arranz; **S. M. Arseneau** et al., 2025, *Sloan Digital Sky Survey-V: Pioneering Panoptic Spectroscopy,* The Astronomical Journal, 171, 52 (arXiv:2507.06989) [ADS Link]
5. SDSS Collaboration, G.A. Pallathadka, M. Aghakhanloo, J. Aird, A. Almeida, S. Amrita, F. Anders, S.F. Anderson, **S. Arseneau** et al., 2025, *The Nineteenth Data Release of the Sloan Digital Sky Survey,* The Astrophysical Journal, In Press (arXiv:2507.07093) [ADS Link]
4. N.R. Crumpler; V. Chandra; N.L. Zakamska; G.A. Pallathadka; **S. Arseneau**; N. Gentile Fusillo; J.J. Hermes; C. Badenes; P. Chakraborty; B.T. Gänsicke; S.P. Schmidt, 2024, *Detection of the Temperature Dependence of the White Dwarf Mass-Radius Relation with Gravitational Redshifts,* The Astrophysical Journal, 977 2 (arXiv:2412.14331) [ADS Link]

3. G.A. Pallathadka; V. Chandra; N.L. Zakamska; H-C. Hwang; Y. Zentai; J.J. Hermes; K. El-Badry; B.T. Gänsicke; S. Morrison; N.R. Crumpler; **S. Arseneau**, 2024, *Discovery of A Proto-White Dwarf With A Massive Unseen Companion,* Astrophysical Journal, 968 1 (arXiv:2310.16313) [ADS Link]

2. A. Barlis; **S. Arseneau**; C.L. Bennett; T. Essinger-Hileman; H. Guo; K.R. Helson; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J.Wollack, 2022, *Characterization of aerogel scattering filters for astronomical telescopes,* Proc. SPIE 12190, Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI, 121902I (arXiv:2208.04257) [ADS Link]

1. K.R. Helson; **S. Arseneau**; A. Barlis; C.L. Bennett; T.M. Essinger-Hileman; H. Guo; T. Marriage; M.A. Quijada; A.E. Tokarz; S.L. Vivod; E.J. Wollack, 2022, *Novel infrared-blocking aerogel scattering filters and their applications in astrophysical and planetary science,* Proc. SPIE 12190, Millimeter, Submillimeter, and Far-Infrared Detectors and Instrumentation for Astronomy XI, 121901P (arXiv:2208.03755) [ADS Link]

Non-Peer-Reviewed Publications —

2. G.A. Pallathadka et. al, incl. **S. M. Arseneau**; 2026. Constraints on the Double White Dwarf Binary Population from SDSS-V DR19. American Astronomical Society Meeting Abstracts, 247, 428.01. [ADS Link]

1. G.A. Pallathadka et. al, incl. **S. M. Arseneau**; 2025. Double Degenerate WD Binaries in SDSS-V DR19. American Astronomical Society Meeting Abstracts, 245, 257.04. [ADS Link]

Professional References

- J. J. Hermes (Boston University, PhD Advisor) jjhermes@bu.edu
- Nadia L. Zakamska (Johns Hopkins University, Advisor) zakamska@jhu.edu
- Carles Badenes (University of Pittsburgh) badenes@pitt.edu